

Data and Moments

Companion Specification for the Segmented Search Model

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1 Purpose of this document

This document specifies the data inputs, the targeted moments, and the moment-to-parameter mapping used to estimate the segmented search model. It is a companion to the model notes: the model notes define the equilibrium objects and the parameters; this document defines what is measured in the data, how each empirical moment is constructed, what it identifies, and how the empirical moments are combined into the SMM objective.

It follows the structure: first the data sources and what we extract from each; then the moments and their identification role; then the estimation windows; then construction recipes; then the weight matrix, influence functions, and file layout; finally, the externally set parameters that are not part of the SMM loop.

Convention. *Unskilled*: no completed four-year degree. *Skilled*: bachelor’s degree or higher. Formally, $\text{skilled}_i = \mathbf{1}\{\text{EDUC}_i \geq 111\}$ (IPUMS code 111 = Bachelor’s degree).

Notation. w_i denotes the survey weight (WTFINL_i for CPS Basic, ASECWT_i for ASEC). $\text{LF}_i = \mathbf{1}\{\text{LABFORCE}_i = 2\}$; $\text{NILF}_i = 1 - \text{LF}_i$. $\text{train}_i = \mathbf{1}\{\text{SCHLCOLL}_i \in \{3, 4\} \text{ and } \text{EDUC}_i < 111\}$ (enrolled in college, no BA). $S_i = \mathbf{1}\{\text{EDUC}_i \geq 111\}$. Model objects: agg_uU , agg_eU are the unskilled unemployed/employed masses; agg_uS , agg_eS the skilled equivalents; agg_t the training mass; $\text{agg_mS} = \text{agg_uS} + \text{agg_eS}$; $\text{model_lf} = \text{agg_uU} + \text{agg_eU} + \text{agg_mS}$ (excludes the training mass); $\text{total_pop} = \text{model_lf} + \text{agg_t} = 1$.

Model variant covered. The estimation uses the variant with $\xi_S = 0$ (all skilled separations are endogenous via $\lambda_S \Gamma(p_S^*)$) and a discrete cross-market search policy $d(x) \in \{0, 1\}$ for skilled-unemployed workers. In stationary equilibrium $d(x) \equiv 0$; the cross-market channel activates only along the simulated transition between equilibria. The SMM objective and the solver inside it work in stationary equilibrium, so the data-side construction proceeds under the convention $\text{EDUC} \geq 111 \Leftrightarrow \text{trained}$.

Part I Data Sources

2 Current Population Survey — Basic Monthly

What it gives us. Unemployment rates, the skilled share, the training share, all UE / EU transition rates, the training-entry hazard `train_entry_rate_U`, and the industry composition needed for the JOLTS vacancy allocation. The AGE variable also supports the life-table pre-estimation of ν .

Source. IPUMS-CPS Basic Monthly extracts (<https://cps.ipums.org>).

Extract window. January 2000 – December 2022 (with a one-month buffer for panel matching). This covers all four estimation windows defined in Part III.

Required variables.

Panel matching for transitions. The CPS uses a 4-8-4 rotation. Match on `CPSIDP` across consecutive months within a rotation spell. Valid month-pairs: `MISH` $\in \{1, 2, 3, 5, 6, 7\}$ matched to the following month. The same matched panel is used for the four CPS transition moments (`jfr_j`, `sep_rate_j`) and for `train_entry_rate_U`.

Table 1: CPS Basic Monthly — required variables.

IPUMS name	Type	Use
YEAR	id	Calendar year
MONTH	id	Calendar month
CPSID	id	Household linking key
CPSIDP	id	Person linking key (cross-month matching)
MISH	id	Month-in-sample (rotation group 1–8)
WTFINL	wt	Basic Monthly final weight
EMPSTAT	stat	Employment status (employed / unemployed / NILF)
LABFORCE	stat	In labour force (yes/no)
WHYABSNT	stat	Reason for absence (COVID misclassification correction)
EDUC	demog	Educational attainment (detailed). Skill indicator: $S = 1$ iff $EDUC \geq 111$.
SCHLCOLL	demog	Currently enrolled in school/college
AGE	demog	Age (restrict to 16–64; used in life-table ν)
SEX	demog	Sex
IND	occ	Industry (Census codes $\rightarrow$ 2-digit NAICS). Used for JOLTS vacancy allocation.
CLASSWKR	stat	Class of worker (wage/salary vs. self-employed)

Industry-to-supersector mapping. For the JOLTS vacancy allocation we map CPS IND (Census industry codes) to the 17 JOLTS supersectors used as the allocation basis (Section 4). The mapping is verified against the 2024 ACS code list and constructed so that each Census code routes to exactly one allocation supersector or is excluded (agriculture, armed forces, private households).

COVID-era correction. During March–June 2020, the CPS suffered response-rate drops and a well-documented misclassification of temporarily laid-off workers as “employed, absent from work.” We apply the BLS correction: for the March–June 2020 months, persons coded as employed with WHYABSNT = 15 (“other reason for absence”) are reclassified to EMPSTAT = 21 (unemployed on temporary layoff). This correction is applied before any moments are computed.

Codes used in the pipeline.

- *Employed*: EMPSTAT $\in \{10, 12\}$.
- *Unemployed*: EMPSTAT $\in \{20, 21, 22\}$.
- *NILF*: EMPSTAT $\in \{30, 31, 32, 33, 34, 36\}$.
- *Civilian labour force*: LABFORCE = 2.
- *Wage/salary worker*: CLASSWKR $\in \{22, 25, 27, 28\}$.
- *Enrolled in college without BA (the train flag)*: SCHLCOLL $\in \{3, 4\}$ (full-time or part-time college) and EDUC < 111.

3 CPS Annual Social and Economic Supplement (ASEC)

What it gives us. Wage moments: median real hourly wage by skill group, wage dispersion (variance), and the log skill premium.

Source. IPUMS-CPS ASEC samples.

Extract window. ASEC 2004–2023 (covering income earned in 2003–2022). This range covers all four estimation windows.

Table 2: CPS ASEC — required variables.

IPUMS name	Type	Use
YEAR	id	Survey year
CPSID	id	Household linking
CPSIDP	id	Person linking
ASECWT	wt	ASEC supplement weight
AGE	demog	Age (restrict 16–64)
SEX	demog	Sex
EDUC	demog	Educational attainment
EMPSTAT	stat	Employment status
INCWAGE	income	Annual wage/salary income
WKSWORK1	income	Weeks worked last year (continuous)
UHRSWORKLY	income	Usual hours per week last year
CLASSWKR	stat	Class of worker
CPI99	price	CPI-U deflator (or deflate with BLS CPI-U-RS)

Required variables.

Hourly wage construction.

$$w_{it} = \frac{\text{INCWAGE}_{it}}{\text{WKSWORK1}_{it} \times \text{UHRSWORKLY}_{it}} \times \frac{\text{CPI-U-RS}_{2012}}{\text{CPI-U-RS}_t}. \quad (1)$$

Trim at the 1st and 99th percentiles of the pooled within-window distribution.

Sample restrictions (ASEC).

1. Age 16–64 at survey date.
2. Employed wage/salary workers (CLASSWKR).
3. Positive wage income, weeks, and hours.
4. Exclude self-employed and unpaid family workers.

4 JOLTS

What it gives us. Vacancy counts by industry, used together with CPS-based unemployment by skill to construct skilled and unskilled market tightness.

Source. BLS JOLTS flat files (job openings level, monthly, seasonally adjusted).

Extract window. December 2000 – December 2022. Covers the longest estimation window (crisis_covid) through 2022-12.

Required series. All series are seasonally adjusted levels (SA, level). The 17 industry series below are mutually exclusive and collectively exhaustive over nonfarm employment; they are used for the skill-share allocation. Agriculture (NAICS 11) is excluded because JOLTS covers nonfarm payrolls only. The six aggregate series are downloaded for cross-checking only and are *never* included in the allocation sum.

Series ID	Supersector	NAICS	Role
<i>Allocation series (sum these)</i>			
JTS1000000000000000JOL	Mining and logging	21	allocation
JTS2300000000000000JOL	Construction	23	allocation
JTS3200000000000000JOL	Durable goods manufacturing	31–33 ^a	allocation
JTS3400000000000000JOL	Nondurable goods manufacturing	31–33 ^a	allocation
JTS4200000000000000JOL	Wholesale trade	42	allocation
JTS4400000000000000JOL	Retail trade	44–45	allocation
JTS4800990000000000JOL	Transportation, warehousing & utilities	22, 48–49	allocation
JTS5100000000000000JOL	Information	51	allocation
JTS5100990000000000JOL	Financial activities	52–53	allocation
JTS5400990000000000JOL	Professional and business services	54–56	allocation
JTS6100000000000000JOL	Educational services	61	allocation
JTS6200000000000000JOL	Health care and social assistance	62	allocation
JTS7100000000000000JOL	Arts, entertainment, and recreation	71	allocation
JTS7200000000000000JOL	Accommodation and food services	72	allocation
JTS8100000000000000JOL	Other services	81	allocation
JTS9100000000000000JOL	Federal government	92	allocation
JTS9200000000000000JOL	State and local government	92	allocation
<i>Aggregate cross-checks only (do not sum)</i>			
JTS0000000000000000JOL	Total nonfarm	—	cross-check
JTS3000000000000000JOL	Manufacturing	31–33	cross-check
JTS4000000000000000JOL	Trade, transport & util.	22, 42, 44–45, 48–49	cross-check
JTS6000000000000000JOL	Education & health	61–62	cross-check
JTS7000000000000000JOL	Leisure & hospitality	71–72	cross-check
JTS9000000000000000JOL	Government	92	cross-check

^a Durable and nondurable goods manufacturing together constitute NAICS 31–33; the two series are mutually exclusive and sum to JTS3000000000000000JOL.

5 Census Job-to-Job Flows (J2J)

What it gives us. The skilled employer-to-employer rate `ee_rate_S`. J2J is preferred to the CPS employer-change question because (i) it has full coverage from 2000 onward, (ii) the employer change is identified from administrative LEHD records rather than a self-reported survey item, and (iii) it provides counts by education group (E0–E4) at quarterly frequency.

Source. Census Bureau J2J Origin-Destination flows, national-level public-use release (<https://lehd.ces.census.gov/data/j2j/>).

Extract window. 2000Q2 – 2022Q4 (covers the longest estimation window).

Series selection. Filter the national release with the following exact conditions, applied identically across all four windows:

- Seasonally adjusted flag `seasonadj = S`;
- Geography `geo_level = N` (national);
- Industry `industry = 00` (all);
- All demographic margins set to “all”: `sex = 0`, `agegrp = A00`, `race = A0`, `ethnicity = A0`, `firmage = 0`, `firmsize = 0`;
- Education `education = E4` (Bachelor’s degree or higher) at the origin firm.

The relevant magnitudes are:

- **EEHire**: count of EE hires from origin firm A to destination firm B in quarter t (the J2J flow).
- **MainB**: stock of main-job-holding workers at firm B .

Quarterly EE rate and monthly conversion. For each retained row in window w the quarterly hazard is

$$ee_t^Q = \frac{EEHire_t}{MainB_t},$$

and the implied constant-hazard monthly rate is

$$ee_t^M = 1 - (1 - ee_t^Q)^{1/3}.$$

`ee_rate_S` is the unweighted mean of ee_t^M over the quarters falling within window w . If the E4 cell is missing in any quarter, that quarter is dropped and the omission is recorded in the diagnostics. There is no fallback to all-workers cells.

6 National Student Clearinghouse (NSC) + IPEDS

What it gives us. Two products from the same NSC file: (i) the ingredients for the training completion rate ϕ — average enrollment in 4-year and 2-year institutions (E_{4yr} , E_{2yr}), pooled across all available Fall semesters, plus the corresponding median time-to-degree (d_{4yr} , d_{2yr}); and (ii) the per-window NSC IPEDS-Universe total enrolment count that anchors the `training_share` level adjustment κ_w (Sections 10 and 15). NSC is used only as a deterministic level reference; it is not introduced as its own moment in the SMM vector (NSC has 3–5 Fall observations per window, which is too few to support an influence-function variance estimate).

Source. NSC Enrollment Coverage Report (annual, Excel), which provides IPEDS Universe enrollment counts for 4-year and 2-year institutions at the US level for each Fall semester. The file covers Fall 2003 – Fall 2024.

Sheet layout. The relevant sheet has rows indexed by (*State or Region, Institution Sector*) and one column per Fall semester. Column headers vary across years (e.g. `Fall_2003` in the earlier years, `Fall_2017` in later years, and `Fall_2024*` for preliminary estimates); the parser accepts all three formats. The US-level rows for “All 4-year Institutions” and “All 2-year Institutions” supply E_{4yr} and E_{2yr} . The US-level row for “Overall” supplies the NSC IPEDS-Universe total count entering κ_w .

Time-to-degree values. NCES median time-to-degree

$$d_{4yr} = 49 \text{ months}, \quad d_{2yr} = 37 \text{ months}.$$

The 49-month figure for bachelor’s programmes captures the typical 4-year programme plus a small completion-lag premium; the 37-month figure for associate’s programmes likewise exceeds the nominal two-year length.

Why two programme types. Both 4-year (bachelor’s) and 2-year (associate’s) programmes constitute “training” in the model, since ϕ governs aggregate transition from training to skilled employment. Treating ϕ as the enrollment-weighted average across the two completion technologies (rather than picking one) avoids under-counting completion among workers in shorter programmes.

Part II Moments and Identification

7 Moment catalogue

Table 3 lists all 18 targeted moments. Each is computed separately for every estimation window.

Table 3: Targeted moments.

Code name	Block	Definition
ur_total	Stocks	Overall unemployment rate, on the training-excluded LF denominator: $(U_U + U_S)/(N_U + N_S)$ with $\text{train}_i = 0$ enforced on the denominator.
ur_U	Stocks	Unskilled unemployment rate: U_U/N_U , denominator restricted to $\text{LF} \cap \neg \text{train}$.
ur_S	Stocks	Skilled unemployment rate: U_S/N_S .
skilled_share	Stocks	Skilled share of training-excluded LF: $N_S/(N_U + N_S)$ with $\text{train}_i = 0$ enforced on the denominator.
training_share	Stocks	Full-time students (enrolled in college without BA and NILF) as a fraction of working-age (16–64) population. Level is rescaled to the NSC IPEDS-Universe anchor via κ_w (Section 10).
emp_var_U	Stocks	Variance of real hourly wages, employed unskilled.
emp_var_S	Stocks	Variance of real hourly wages, employed skilled.
jfr_U	Trans	Monthly unskilled job-finding rate: $\Pr(E_{t+1} \mid U_t, U)$.
sep_rate_U	Trans	Monthly unskilled EU separation rate: $\Pr(U_{t+1} \mid E_t, U)$.
jfr_S	Trans	Monthly skilled job-finding rate: $\Pr(E_{t+1} \mid U_t, S)$.
sep_rate_S	Trans	Monthly skilled EU separation rate: $\Pr(U_{t+1} \mid E_t, S)$.
ee_rate_S	Trans	Monthly skilled employer-to-employer rate (Census J2J, E4 only).
train_entry_rate_U	Trans	Monthly hazard of an unskilled, unemployed, not-currently-enrolled worker becoming enrolled in the following month.
p50_wage_U	Wages	Median real hourly wage, employed unskilled (normalised).

Code name	Block	Definition
p50_wage_S	Wages	Median real hourly wage, employed skilled (normalised).
wage_premium	Wages	Log skill premium: $E[\log w \mid S] - E[\log w \mid U]$.
theta_U	Tight	Unskilled V-U ratio: $\theta_U = V_U/U_U$.
theta_S	Tight	Skilled V-U ratio: $\theta_S = V_S/U_S$.

8 Identification

This section explains *what each moment identifies* in the model: which parameters each moment primarily constrains, and why the set as a whole is sufficient.

8.1 Stock moments

- **ur_U, ur_S**: pin down the balance between job-finding and separation rates in each segment. Given tightness, they constrain matching efficiencies and shock hazards.
- **skilled_share**: constrains the training cutoff $\bar{x}$ at the long-run-stock margin and therefore the interaction between the type distribution (a_ℓ, b_ℓ) and the training cost coefficient c .
- **training_share**: identifies the mass in the training state, hence the cutoff $\bar{x}$ conditional on the (pre-estimated) training-completion rate ϕ .
- **emp_var_U**: variance of the unskilled wage distribution. Because the unskilled wage is linear in match quality p for fixed x (see model notes), the variance of wages maps into the shape of the match-quality distribution $G(p) = p^{\alpha_U}$, identifying the damage-shock shape α_U jointly with the reservation cutoff p^* (which is pinned independently by **sep_rate_U**).
- **emp_var_S**: analogous role for the skilled block. Constrains the offer distribution $\Gamma = \text{Beta}(a_\Gamma, b_\Gamma)$, modulated by the skill-bias exponent γ_{P_S} via the convex transformation $x \mapsto x^{\gamma_{P_S}}$.

8.2 Transition rates

- **jfr_U**: in the model the unskilled job-finding rate equals the contact rate $f_U = \mu_U \theta_U^{1-\eta_U}$, because all new unskilled matches start at frontier efficiency $p_0 = 1 > p^*(x)$, so acceptance is automatic. Given θ_U from JOLTS, **jfr_U** identifies the regime-specific unskilled matching parameters (μ_U, η_U) .
- **jfr_S**: the skilled job-finding rate is the acceptance-weighted contact rate. Because a skilled offer is a draw $\tilde{p} \sim \Gamma$ and the worker rejects whenever $\tilde{p} < p_S^*(x)$, the model object is

$$\text{jfr}_S^{\text{model}} = \frac{\kappa_S \int (1 - \Gamma(p_S^*(x))) u_S(x) dx}{\int u_S(x) dx}, \quad \kappa_S = \mu_S \theta_S^{1-\eta_S}.$$

Given θ_S it identifies (μ_S, η_S) jointly with the lower tail of the offer distribution.

- **sep_rate_U, sep_rate_S**: the EU flow. In the unskilled block the model object is the employment-weighted endogenous destruction hazard

$$\text{sep_rate}_U^{\text{model}} = \frac{\lambda_U \int G(p^*(x)) e_U(x) dx}{\int e_U(x) dx}, \quad G(p) = p^{\alpha_U}.$$

In the skilled block, with $\xi_S = 0$, the model object is

$$\text{sep_rate}_S^{\text{model}} = \frac{\lambda_S \int \Gamma(p_S^*(x)) e_S^{\text{tot}}(x) dx}{\int e_S^{\text{tot}}(x) dx}.$$

Demographic exits (rate ν) appear on neither side: in the data they map to E→NILF transitions, in the model they exit the population. Jointly with the wage-variance moments, these moments identify the shock arrival rates and the reservation cutoffs.

- **ee_rate_S**: the skilled job-to-job rate. In the model this is the OJS quit hazard

$$\text{ee_rate_S}^{\text{model}} = \frac{\int s^*(x, p) \kappa_S (1 - \Gamma(p)) e_S(x, p) dx dp}{\int e_S^{\text{tot}}(x) dx},$$

where $s^*(x, p) = \mathbf{1}\{p < p_S^{\text{oj}}(x)\}$ is the on-the-job search policy. This is the key moment separating the offer distribution Γ from the separation cutoff p_S^* , and also pins down the OJS cost σ_S (since workers search only if the ladder gain exceeds σ_S).

- **train_entry_rate_U**: in the model, training is initiated by an unskilled-unemployed worker of type x whenever $x > \bar{x}$, with $\bar{x}$ the training cutoff. The stationary model counterpart is

$$\text{train_entry_rate_U}^{\text{model}} = \frac{\int_{\bar{x}}^1 u_U(x) dx}{\text{agg_uU}}.$$

This is a flow object that pins down $\bar{x}$ at the entry margin *separately* from ϕ and from the slow stock adjustment of the trained mass. Regime variation in **train_entry_rate_U** between **base_covid** and **crisis_covid** identifies whether a COVID-era training decline reflects a rise in the training cost c (a structural shift in the decision rule) or a compositional change in the unskilled-unemployed pool (which moves the stock but not the entry hazard).

8.3 Wages

- **p50_wage_U**, **p50_wage_S**: median real hourly wages by skill group. Wage levels identify the productivity shifters P_U and γ_{P_S} given the regime-specific bargaining weights β_j . The median is preferred to the mean because it is less sensitive to the upper tail of the (winsorised) ASEC wage distribution.
- **wage_premium**: the log skill premium is the primary moment identifying the *return to training*. It constrains γ_{P_S} relative to P_U and the value of the training option. Invariant to the within-window wage normalisation, so it is the cleanest of the wage moments at the model-data interface.
- **emp_var_U**, **emp_var_S**: wage dispersion identifies the regime-specific bargaining weights β_j . Higher bargaining power compresses wages; lower power allows more pass-through of match-quality heterogeneity. (Listed in the stocks block of the catalogue because they are within-period second moments of the wage distribution rather than transitions.)

8.4 Tightness

- **theta_U**, **theta_S**: directly constrain regime-specific vacancy costs k_U, k_S through free entry. Given matching parameters and the composition of unemployed workers, free entry pins down tightness uniquely.

8.5 Deep vs. regime-specific identification

Under the current parameter classification, the deep set is $\{a_\ell, b_\ell, b_U, b_T, b_S\}$. These parameters are identified primarily from stock moments and the skill premium, which respond slowly across regimes:

- The type distribution (a_ℓ, b_ℓ) is identified primarily from **skilled_share** and **wage_premium**, which together pin down the location and dispersion of innate ability.

- Flow payoffs b_U, b_T, b_S are identified from the level of unemployment values and training incentives, manifested in unemployment rates, the skilled share, and the position of wage distributions near the reservation cutoffs.

All other parameters — the training cost coefficient c , the matching parameters (η_j, μ_j) , the bargaining weights β_j , the OJS cost σ_S , the productivity shifters (P_U, γ_{P_S}) , the vacancy costs (k_U, k_S) , the shock hazards (λ_j) , and the match-quality shape parameters $(\alpha_U, a_\Gamma, b_\Gamma)$ — are regime-specific and identified *within each window*:

- Training cost coefficient c : from `train_entry_rate_U` (flow margin) and `training_share` (stock), in conjunction with `skilled_share`, since c governs the level of the training cutoff $\bar{x}$. The flow margin separates a change in c from composition shifts in the unskilled-unemployed pool.
- Matching parameters (η_j, μ_j) : from the joint level of `jfr_j` and `theta_j` within window.
- Bargaining weights β_j : from wage dispersion (`emp_var_j`) within window.
- OJS cost σ_S : from `ee_rate_S` jointly with the wage moments, since the OJS cutoff p_S^{oj} responds to σ_S holding match-quality shape fixed.
- Productivity shifters and vacancy costs: from wage levels and tightness levels.
- Shock arrival rates and damage / offer shapes: from separation rates and wage-variance shifts.

Testable implication: did matching technology shift? A pure aggregate-demand shock with no technology change predicts that the matching elasticities, efficiencies, bargaining weights, and OJS cost are *approximately equal* across the base→crisis pair within a single crisis episode. A supply / reallocation shock that re-curves the Beveridge curve permits these parameters to differ. The two crisis pairs in this paper deliver opposite verdicts on this test: the FC pair is consistent with stable matching and bargaining technology (a textbook ride along the Beveridge curve), whereas the COVID pair shows a large shift in separation rates at nearly unchanged tightness, which can only be rationalised by a change in some subset of $(\mu_j, \eta_j, \beta_j, \sigma_S, \lambda_j, \alpha_U, \Gamma)$. The multi-regime estimation recovers the answer.

Part III Estimation Windows

9 The four windows

The multi-regime estimation requires moments computed over four distinct periods, each approximating a stationary regime:

Table 4: Estimation windows.

Label	Code	Period	Role
Pre-FC baseline	<code>base_fc</code>	Jan 2003 – Dec 2007	Stage 1: estimate all parameters (60 months).
Financial crisis	<code>crisis_fc</code>	Jan 2008 – Dec 2010	Stage 2a: regime-specific only (36 months).
Pre-COVID baseline	<code>base_covid</code>	Jan 2015 – Dec 2019	Structural stability check + 2nd baseline (60 months).
COVID crisis	<code>crisis_covid</code>	Jan 2020 – Dec 2022	Stage 2b: regime-specific only (36 months).

Each window produces 18 moment values plus an 18×18 variance-covariance matrix from

influence functions. The two baseline windows are 60 months and the two crisis windows are 36 months, by symmetric design.

9.1 Common sample restrictions

1. Age 16–64, civilian labour force.
2. Exclude armed forces.
3. Wage moments: wage/salary workers, positive income/weeks/hours, exclude self-employed.
4. Transition rates: matched month-pairs with valid CPSIDP, MISH $\in \{1, 2, 3, 5, 6, 7\}$.

9.2 Window-specific notes

base_fc (Jan 2003 – Dec 2007). Starts roughly a year after the 2002 trough of the post-dot-com recession. NBER expansion: Nov 2001 – Dec 2007.

crisis_fc (Jan 2008 – Dec 2010). Covers the NBER contraction (Dec 2007 – Jun 2009) and the slow post-trough recovery through 2010, when tightness had fallen sharply and had not yet returned to the pre-FC locus. The window ends before the 2011 fiscal cliff and the slow normalisation that followed.

base_covid (Jan 2015 – Dec 2019). Mature post-crisis expansion. Starting in 2015 avoids the early recovery (2010–2014). Tests structural stability of deep parameters across a decade.

crisis_covid (Jan 2020 – Dec 2022). NBER peak Feb 2020; the contraction was officially short, but the labour market did not return to the pre-COVID locus by end-2022. The window covers the pandemic shock, the rebound, and the sustained outward Beveridge-curve shift through 2022. BLS misclassification correction applied for Mar–Jun 2020.

9.3 Symmetric window design

The two baseline windows are 60 months each and the two crisis windows are 36 months each. This symmetric design lets the SMM treat the two crisis pairs on equal footing. The 36-month length is long enough to span the bulk of each crisis episode plus the immediate adjustment, and short enough to remain a distinguishable regime from the surrounding baseline.

FC: a ride along the Beveridge curve. Between **base_fc** and **crisis_fc** market tightness fell sharply (by roughly a third), unemployment rose, and separation rates rose modestly. The Beveridge curve traced out by the time series over 2008–2010 lies approximately on the same locus as the pre-FC data. This is the textbook signature of a pure aggregate-demand shock with unchanged matching, bargaining, and OJS technology.

COVID: an outward Beveridge-curve shift. Between **base_covid** and **crisis_covid** the Beveridge curve shifted outward: tightness fell only modestly, but separation rates roughly doubled for both skill groups. The pre-shock locus is not recovered by end-2022. This is consistent with a change in some subset of the matching / bargaining / OJS / shock parameters, which the model classifies as regime-specific precisely so that the data can answer the question.

9.4 ASEC year mapping

Since ASEC reports income for the *previous* calendar year, the ASEC surveys used for each window are:

- **base_fc:** ASEC 2004–2008 (income years 2003–2007).

- `crisis_fc`: ASEC 2009–2011 (income years 2008–2010).
- `base_covid`: ASEC 2016–2020 (income years 2015–2019).
- `crisis_covid`: ASEC 2021–2023 (income years 2020–2022).

9.5 Stationarity caveat

The model produces stationary-equilibrium moments. Short crisis windows are not true stationary states. We interpret crisis-window averages as the “quasi-stationary” centre of gravity. In stationary equilibrium the discrete cross-market policy $d(x)$ is identically zero; it activates only along the simulated transition between equilibria.

Part IV Moment Construction Recipes

10 Labour-market stocks

All recipes below are applied identically within each window w . Stock moments are computed by the *per-period-then-average* approach: compute a monthly value first, then average across all months in the window. All four stock denominators that touch the labour force enforce the working-student exclusion $LF_i \wedge \neg \text{train}_i$ so that the data denominator aligns with the model’s `model_lf`, which excludes the training mass.

ur_total. For each month t in window w :

$$\text{ur_total}_t = \frac{\sum_{i: \text{unemp}_t} w_i}{\sum_{i: LF_t \wedge \neg \text{train}_t} w_i}.$$

Average over all months in the window. The denominator excludes working students; the numerator is unaffected because employed absent-from-work students and unemployed students are exclusive categories under the IPUMS coding.

ur_U.

$$\text{ur_U}_t = \frac{\sum_{i: \text{unemp}_t, \neg S} w_i}{\sum_{i: LF_t \wedge \neg \text{train}_t, \neg S} w_i}.$$

Same working-student filter on the denominator as for **ur_total**, restricted to unskilled ($S = 0$).

ur_S.

$$\text{ur_S}_t = \frac{\sum_{i: \text{unemp}_t, S} w_i}{\sum_{i: LF_t, S} w_i}.$$

No working-student filter needed: by construction $\text{train} = 1 \Rightarrow \text{EDUC} < 111$, so the skilled denominator never contains a trainee.

skilled_share. Monthly fraction of the training-excluded LF that is skilled, averaged over the window:

$$\text{skilled_share}_w = \frac{1}{T_w} \sum_{t \in w} \frac{\sum_{i: LF_t, S} w_{i,t}}{\sum_{i: LF_t \wedge \neg \text{train}_t} w_{i,t}}.$$

The denominator carries the working-student filter; the numerator does not (working students are by construction not skilled).

training_share (raw). Strict variant: the numerator is restricted to NILF trainees (full-time students whose state coincides with the model’s training state, which sits outside the labour force), and the denominator is the working-age population (already restricted to 16–64 upstream):

$$\text{training_share}_w = \frac{1}{T_w} \sum_{t \in w} \frac{\sum_{i: \text{train}_t \wedge \text{NILF}_t} w_{i,t}}{\sum_{i: \text{pop}_t} w_{i,t}}.$$

This matches the model definition $\text{training_share} = \text{agg_t}/\text{total_pop}$ with $\text{total_pop} = 1$. The raw level produced by this expression is under-coverage-biased and not cross-window comparable; the next paragraphs introduce the NSC-anchored adjustment that the SMM actually targets.

Universe note for training_share. IPUMS SCHLCOLL is collected only for ages 16–24 in the pre-2013 era and is universe-expanded to ages 16–54 from Jan 2013 onward. Within `base_fc` and `crisis_fc` the universe is consistent (both restrict to 16–24 implicitly through the missing-data structure). Within `base_covid` and `crisis_covid` the universe is also consistent (both include 25+). Cross-FC-vs-COVID comparisons of the raw CPS moment, however, are confounded by the universe change. A direct diagnostic is to tabulate the CPS `in_training` share by age band $\times$ window: the bands 25+ are structurally zero in the FC windows and non-trivial in the COVID windows, precisely because of the SCHLCOLL universe limit.

NSC-based level adjustment (κ_w). To make `training_share` comparable across the FC and COVID windows and to correct its absolute level (the raw CPS construction understates the true postsecondary trainee mass even within each window), we apply a per-window deterministic scale factor calibrated to the NSC IPEDS-Universe Fall headcount. Let $\widehat{\text{enr}}_w^{\text{NSC}}$ be the NSC US-“Overall” IPEDS Universe enrolment, averaged across the Fall semesters falling within window w , and let $\widehat{\text{enr}}_w^{\text{CPS}} = \frac{1}{T_w} \sum_{t \in w} \sum_{i: \text{train}_t \wedge \text{NILF}_t} w_{i,t}$ be the CPS-implied enrolment count (per-month then averaged, identical numerator weighting to the raw moment). Define

$$\kappa_w = \frac{\widehat{\text{enr}}_w^{\text{NSC}}}{\widehat{\text{enr}}_w^{\text{CPS}}}.$$

The adjusted data target consumed by the SMM is

$$\text{training_share}_w^{\text{adj}} = \kappa_w \cdot \text{training_share}_w \equiv \frac{\widehat{\text{enr}}_w^{\text{NSC}}}{\widehat{\text{Pop}}_w^{\text{CPS}}}.$$

The model counterpart $\text{agg_t}/\text{total_pop}$ is age-uncapped by construction and is directly comparable to the adjusted target.

Identifying assumptions for κ_w . The rescaling is exact under three assumptions. Posit a multiplicative coverage model

$$\begin{aligned} \widehat{m}_w^{\text{CPS}} &= \alpha_w \tilde{m}_w (1 + \varepsilon_w^{\text{CPS}}), \\ \widehat{m}_w^{\text{NSC}} &= \tilde{m}_w (1 + \varepsilon_w^{\text{NSC}}), \end{aligned}$$

where $\tilde{m}_w$ is the true age-uncapped postsecondary training share, $\alpha_w < 1$ captures the CPS under-coverage, and the ε ’s are mean-zero noise.

- **A1 (NSC $\approx$ census).** $\text{Var}(\widehat{m}_w^{\text{NSC}}) \approx 0$. NSC is an institutional reporting system, not a survey; the IPEDS Universe coverage rate is published in the Coverage sheet of the same NSC file and runs $\approx 86\%$ in 2003 climbing to $\approx 98\%$ from 2015 onward. Residual non-coverage is institutional (small or specialty schools that don’t report), not random sampling, so it does not contribute a variance term.

- **A2 (α_w is structural within window).** α_w is a fixed property of the CPS measurement instrument in window w , not a behavioural object. Three components support this: (i) the SCHLCOLL age universe is a hard IPUMS feature and changes only in Jan 2013, between the FC and COVID pairs; (ii) the EDUC < 111 filter is definitional; (iii) the NILF intersection is part of the model-data mapping (the model’s training state is defined as NILF), not a measurement choice that worker behaviour can invalidate.
- **A3 (Plug-in for $\hat{\kappa}_w$).** Treat the realised $\hat{\kappa}_w$ as constant. This is the standard two-step estimator move. Here it biases the variance *upward*: the product $\hat{\kappa}_w \cdot \hat{m}_w^{\text{CPS}}$ collapses algebraically to $\widehat{\text{enr}}_w^{\text{NSC}} / \widehat{\text{Pop}}_w^{\text{CPS}}$, and the CPS population total has CV well under 1%, so the true $\text{Var}(\hat{m}_w^{\text{adj}})$ is substantially smaller than $\kappa_w^2 \text{Var}(\hat{m}_w^{\text{CPS}})$. The SMM therefore down-weights `training_share` *more* than the optimal-weight matrix would, which is a loss of efficiency, not a bias.

Under A1–A3, bilinearity gives exactly

$$\text{Var}(\hat{m}_w^{\text{adj}}) = \kappa_w^2 \text{Var}(\hat{m}_w^{\text{CPS}}), \quad \text{Cov}(\hat{m}_w^{\text{adj}}, m_k) = \kappa_w \text{Cov}(\hat{m}_w^{\text{CPS}}, m_k) \quad \forall k.$$

Off-diagonals are *preserved up to κ_w* and are not zeroed, so the full CPS covariance structure of the other 17 moments is unchanged.

Grad-student note. NSC IPEDS-Universe counts all Title-IV postsecondary enrolment, including graduate students, while CPS `in_training` excludes them via `EDUC` ≥ 111 . κ_w therefore absorbs the grad-student share ($\approx 13\%$ of total US postsecondary enrolment) on top of the SCHLCOLL universe correction. The model’s training state, interpreted as “investment in becoming skilled,” covers grad school in the current variant, so this interpretation is consistent with the SMM target.

emp_var_U, emp_var_S. ASECWT-weighted variance of the normalised hourly wage w^{norm} (Section 12) by skill group, computed within each ASEC year and averaged across years in the window.

11 Transition rates

jfr_U, jfr_S. From matched CPS month-pairs $(t, t+1)$ in window w :

$$\text{jfr}_{j,t} = \frac{\sum_{i: U_t \rightarrow E_{t+1}, j} w_{i,t}}{\sum_{i: U_t, j} w_{i,t}}, \quad j \in \{U, S\}.$$

Averaged over all valid month-pairs in the window. Weight: base-month WTFINL.

sep_rate_U, sep_rate_S.

$$\text{sep}_{j,t} = \frac{\sum_{i: E_t \rightarrow U_{t+1}, j} w_{i,t}}{\sum_{i: E_t, j} w_{i,t}}.$$

This is the $E \rightarrow U$ flow only. Job-to-job moves are not separations. $E \rightarrow \text{NILF}$ transitions are also excluded by construction (they map to demographic exit, not unemployment).

ee_rate_S (from J2J, E4 only). Apply the J2J filter from Section 5 restricting to `education` = E4 at origin. For each retained quarter t , the quarterly hazard is $\text{ee}_t^Q = \text{EEHire}_t / \text{MainB}_t$ and the implied monthly hazard is $\text{ee}_t^M = 1 - (1 - \text{ee}_t^Q)^{1/3}$. **ee_rate_S** is the unweighted mean of ee_t^M over the quarters falling within window w .

train_entry_rate_U. From the CPS Basic matched panel, the monthly hazard of an unskilled, unemployed, not-currently-enrolled worker becoming enrolled in the following month is

$$\text{train_entry_rate_U}_t = \frac{\sum_{i: U_t, \neg \text{train}_t, \neg S \rightarrow \text{train}_{t+1}} w_{i,t}}{\sum_{i: U_t, \neg \text{train}_t, \neg S} w_{i,t}}.$$

Same MISH-pair restriction as the other CPS transition moments. Averaged over month-pairs in the window. The denominator conditions on the model-relevant decision margin (unskilled-unemployed, not already enrolled), so universe-coverage shifts in SCHLCOLL apply symmetrically to numerator and denominator within a window.

12 Wages

All from the ASEC. Hourly wages from (1), trimmed at 1st/99th percentiles within the window.

Normalisation. Each window is normalised by its own pooled median hourly wage $\tilde{w}_{50,w}$: $w_{it}^{\text{norm}} = w_{it}/\tilde{w}_{50,w}$.

Within-window aggregation: per-year-then-average. For all wage moments we compute the moment within each ASEC year in the window and then average across years. This puts every ASEC year on equal footing regardless of sample size, and prevents a single large-sample year from dominating.

p50_wage_U, p50_wage_S. ASECWT-weighted median of w^{norm} by skill group, computed within each ASEC year, then averaged across years in the window.

wage_premium. $E[\log w^{\text{norm}} \mid S] - E[\log w^{\text{norm}} \mid U]$, computed within each ASEC year and averaged. Invariant to the within-window normalisation.

Normalisation check at the model-data interface. The data divides by the within-window pooled median; the model solver uses its own internal wage scale. A one-time visual check of the empirical and model-implied wage densities, plotted on a shared axis, is sufficient to verify that the two scales coincide in normalised units.

13 Market tightness

Step 1. Unemployment by skill from CPS Basic: $U_{j,t}$.

Step 2. Vacancies by skill via industry allocation (Section 4). Let $s_{k,w}$ be the within-window weighted skilled share of employment in supersector k (computed from CPS using the industry-to-supersector map, with the working-student exclusion inherited from the **skilled_share** construction). Then

$$V_{S,t} = \sum_k s_{k,w} V_{k,t}, \quad V_{U,t} = \sum_k (1 - s_{k,w}) V_{k,t},$$

summing only over the 17 allocation supersectors (never the six aggregate cross-check series). JOLTS units are scaled by 1000 to match CPS counts.

Step 3. $\theta_{j,t} = V_{j,t}/U_{j,t}$; average over months in the window.

Part V SMM Weights, Influence Functions, and File Layout

14 SMM weight matrix

The SMM objective for window w is

$$Q_w(\theta) = [\hat{m}_w - m(\theta)]' \hat{\Xi}_w [\hat{m}_w - m(\theta)],$$

with $\hat{m}_w$ the vector of $K = 18$ data moments and $\hat{\Xi}_w$ a positive-semidefinite weight matrix. The headline estimation uses identity weights ($\hat{\Xi}_w = I$), equivalent to equal weight on each scale-normalised moment deviation. Two robustness configurations are available without re-running the data pipeline: diagonal weights $\hat{\Xi}_w = \text{diag}(1/\hat{\sigma}_{1,w}^2, \dots, 1/\hat{\sigma}_{K,w}^2)$, and the full optimal weight matrix $\hat{\Xi}_w = \hat{\Sigma}_w^{-1}$ (with off-diagonal shrinkage if the condition number of $\hat{\Sigma}_w$ exceeds a target).

Pipeline / SMM separation of concerns. The pipeline produces and stores the full 18×18 covariance matrix $\hat{\Sigma}_w$ for each window. Weight-matrix construction is handled by the SMM code, which inverts (or diagonalises) $\hat{\Sigma}_w$ at load time. Keeping the full $\hat{\Sigma}_w$ on disk means the weight scheme can be changed without re-running the pipeline.

The same separation holds for the κ_w rescaling of **training_share**: the pipeline writes the raw moment and the raw $\hat{\Sigma}_w$; the SMM driver applies κ_w at load time (see Section 15).

15 Influence functions

Each moment $\hat{m}_k = \frac{1}{N} \sum_i \psi_k(z_i) + o_p(N^{-1/2})$. The variance-covariance matrix is $\hat{\Sigma}_w = \frac{1}{N_w} \sum_i \psi(z_i) \psi(z_i)'$.

Sample mean. $\psi_i = y_i - \hat{m}$.

Sample variance. $\psi_i = (y_i - \bar{y})^2 - \hat{\sigma}^2$.

Ratio / rate. $\psi_i = (y_i - \hat{r} x_i)/\bar{x}$. This covers **ur_j**, **skilled_share**, **training_share**, all transition rates including **train_entry_rate_U**, **ee_rate_S**, and **theta_j**.

Median / quantile. $\psi_i = (\mathbf{1}\{y_i \leq \hat{q}_\alpha\} - \alpha)/f(\hat{q}_\alpha)$ for the α -quantile.

Implementation. For each window: (i) compute $\hat{m}_w$; (ii) compute $\psi(z_i)$ for each observation; (iii) form $\hat{\Sigma}_w$; (iv) store both $\hat{m}_w$ and $\hat{\Sigma}_w$. When moments are aggregated across periods (e.g. ASEC year-then-window or CPS month-then-window), the influence-function contributions are stacked at the appropriate aggregation level before forming the outer product, so that within-period clustering is respected.

training_share rescaling at SMM load time. The pipeline stores the raw $\hat{\Sigma}_w$. When the SMM driver loads the matrix it multiplies the row and column corresponding to **training_share**

by κ_w (so the diagonal cell is multiplied by κ_w^2 and every off-diagonal entry in that row/column is multiplied by κ_w). Algebraically:

$$\hat{\Sigma}_w^{\text{adj}}[k, k'] = \begin{cases} \kappa_w^2 \hat{\Sigma}_w[k, k] & \text{if } k = k' = \text{training_share}, \\ \kappa_w \hat{\Sigma}_w[k, k'] & \text{if exactly one of } k, k' = \text{training_share}, \\ \hat{\Sigma}_w[k, k'] & \text{otherwise.} \end{cases}$$

This is exact under A1–A3 of Section 10. All other rows and columns are unchanged; the full CPS covariance structure of the other 17 moments is preserved.

16 File and directory layout

```
project_root/
  code/
    solver/                                # model solver
      model_main.jl
      params.jl, grids.jl, unskilled.jl, skilled.jl
      solver.jl, equilibrium.jl, plots.jl
    smm/                                    # SMM estimation
      smm_main.jl                          # driver
      moments.jl                          # data + model moment loaders;
                                          #   applies kappa_w to training_share
                                          #   target, Sigma row/col, and sigma vec
      smm_params.jl                       # SMMSpec, SMMRunParams, free-param list
      smm.jl                              # SMM objective, DE, SA, NM, multistart
    notebooks/
      data_pipeline.ipynb                 # monolithic data pipeline:
                                          #   cleaning, transitions, J2J, nu, phi,
                                          #   kappa cell, moments, Sigma, validation
  data/
    raw/
      cps_basic/                          # IPUMS extract
      cps_asec/                          # IPUMS ASEC extract
      jolts/                             # BLS flat files
      j2j/                               # Census J2J flows
      nsc/                               # NSC Enrollment-Coverage Excel
    derived/
      windows.json                       # WINDOWS dict; single source of truth
      cps_basic_clean.arrow
      cps_asec_clean.arrow
      transitions_monthly.arrow
      jolts_clean.arrow
      phi_calibration.csv                 # single phi pooled across Fall semesters
      nu_estimation.csv                  # life-table nu on base_fc and base_covid
      training_share_scale.csv           # kappa_w per window (NSC/CPS level anchor)
      cps_vs_nsc_enrollment.csv          # diagnostic table backing kappa_w
      moments_base_fc.csv                # 18 moments per window
      moments_crisis_fc.csv
      moments_base_covid.csv
      moments_crisis_covid.csv
      sigma_base_fc.csv                  # full 18x18 var-cov (raw; SMM applies kappa)
```

```

sigma_crisis_fc.csv
sigma_base_covid.csv
sigma_crisis_covid.csv
output/
plots/
tables/
smm/                                     # serialised SMMResult bundles (.jls)

```

17 Summary: moment-to-data mapping

Table 5: Moment-to-data mapping.

Moment	Source	Key variables
ur_total	CPS Basic	EMPSTAT, LABFORCE, SCHLCOLL, EDUC, WTFINL
ur_U	CPS Basic	same
ur_S	CPS Basic	EMPSTAT, LABFORCE, EDUC, WTFINL
skilled_share	CPS Basic	LABFORCE, SCHLCOLL, EDUC, WTFINL
training_share	CPS Basic + NSC	LABFORCE, SCHLCOLL, EDUC, AGE, WTFINL; NSC IPEDS-Universe Fall headcount for κ_w
emp_var_U	CPS ASEC	INCWAGE, WKSWORK1, UHRSWORKLY, EDUC
emp_var_S	CPS ASEC	same
jfr_U	CPS Basic (matched)	CPSIDP, EMPSTAT, EDUC
jfr_S	CPS Basic (matched)	same
sep_rate_U	CPS Basic (matched)	same
sep_rate_S	CPS Basic (matched)	same
ee_rate_S	Census J2J	EEHire, MainB, education = E4
train_entry_rate_U	CPS Basic (matched)	CPSIDP, EMPSTAT, SCHLCOLL, EDUC
p50_wage_U	CPS ASEC	INCWAGE, WKSWORK1, UHRSWORKLY, EDUC
p50_wage_S	CPS ASEC	same
wage_premium	CPS ASEC	same
theta_U	JOLTS + CPS	openings by industry, IND, EDUC
theta_S	JOLTS + CPS	same

Part VI Externally Set Parameters

18 Overview

Three parameters are set outside the SMM estimation loop. One is *calibrated* to a standard external value; two are *pre-estimated* from external micro-data. r and ϕ are common across all four windows; ν takes two values, one per crisis pair.

19 Calibrated parameter: discount rate r

The discount rate enters every value function as a level shifter: for a generic HJB of the form $(r + \nu + \dots)V = \dots$, a higher r uniformly lowers continuation values. This effect is

Table 6: Externally set parameters.

Symbol	Name	Value	Frequency	Source / method
<i>Calibrated, common across windows</i>				
r	Discount rate	0.0033	monthly	4% annual real rate (risk-free rate)
<i>Pre-estimated from data, common across windows</i>				
ϕ	Training completion rate	— (from data)	monthly	NSC/IPEDS Universe enrollment pooled across Fall semesters 2003–2024 + NCES median time-to-degree; enrollment-weighted across 4-year and 2-year programmes.
<i>Pre-estimated from data, per baseline window</i>				
ν_w	Demographic turnover	— (from data)	monthly	Life-table on CPS Basic LF (ages 16–64), $\hat{\nu}_w = 1/\text{remaining months}$, computed on <code>base_fc</code> and <code>base_covid</code> . FC pair uses $\hat{\nu}_{\text{base_fc}}$; COVID pair uses $\hat{\nu}_{\text{base_covid}}$.

almost perfectly collinear with changes in the flow payoffs b_U, b_T, b_S and the productivity shifters P_U, γ_{P_S} : raising r by ϵ and simultaneously adjusting flow payoffs or productivity can leave all observable moments nearly unchanged. Formally, the Jacobian $\partial m / \partial r$ (evaluated at the baseline) is close to a linear combination of $\partial m / \partial b_U$, $\partial m / \partial b_S$, and $\partial m / \partial P_U$, making the Hessian of the SMM objective near-singular when r is included. The discount rate has no moment that pins it down independently of flow payoffs. We calibrate $r = 0.0033$ monthly ($\approx 4\%$ annual), the standard convention.

20 Pre-estimated parameter: training completion rate ϕ

20.1 Why ϕ is not in the SMM objective

In principle, ϕ is jointly identified by the training share and the training flow rate. In steady state, $\text{training_share} \approx \text{training_flow} / (\phi + \nu)$, so given the two moments one could back out ϕ . In practice the training flow is poorly measured at monthly frequency in the CPS (small enrollment-state transitions are dominated by sampling noise), and small measurement error in the training share translates into large error in ϕ because the training share itself is small ($\approx 2\%$). More fundamentally, ϕ interacts tightly with the training cutoff $\bar{x}$: a model with high ϕ (fast completion) and a low cutoff (few entrants) can produce similar training stocks and flows as a model with low ϕ and a higher cutoff. Without direct microdata on training-spell durations, the two are not separately identified from CPS moments alone. We compute ϕ directly from administrative data on enrollment and time-to-degree.

20.2 Construction recipe

Data inputs. Two ingredients, pooled across all available Fall semesters:

1. Average IPEDS Universe enrollment in 4-year institutions ($E_{4\text{yr}}$) and in 2-year institutions ($E_{2\text{yr}}$), from the NSC Enrollment Coverage Report. For each available Fall semester between 2003 and 2024 we read the US-level rows for “All 4-year Institutions” and “All 2-year Institutions” and take the unweighted mean across years.
2. NCES median time-to-degree: $d_{4\text{yr}} = 49$ months (bachelor’s), $d_{2\text{yr}} = 37$ months (associate’s).

Formula. The enrollment-weighted average completion hazard across the two programme types is

$$\hat{\phi} = \frac{E_{4\text{yr}}(1/d_{4\text{yr}}) + E_{2\text{yr}}(1/d_{2\text{yr}})}{E_{4\text{yr}} + E_{2\text{yr}}}. \quad (2)$$

This is the monthly completion probability for an average enrollee. The same $\hat{\phi}$ is used in all four windows.

21 Pre-estimated parameter: demographic turnover ν_w

We estimate ν separately on each baseline window (`base_fc` and `base_covid`) using the life-table method: for each CPS Basic LF participant of age a , the implied remaining working life is $\max(65 - a, 0) \cdot 12$ months, and $\hat{\nu}_w$ is the reciprocal of its weighted mean over window w .

21.1 Why ν cannot be included in the SMM objective

The turnover rate ν appears in the model symmetrically with r : the effective discount rate in most HJBs is $r + \nu$. As a result, ν suffers from the same collinearity: a change in ν can be offset by adjusting r , leaving value functions and hence observable moments essentially unchanged. Beyond this collinearity, ν also governs the inflow rate of new workers (with density $\ell(x)$), which affects the stationary distributions. The distribution moments (`skilled_share`, `training_share`) are primarily driven by the training cutoff $\bar{x}$ and the type distribution (a_ℓ, b_ℓ) , so the marginal effect of ν on these moments is swamped by the effects of the deep parameters the estimation already targets. Pre-estimating ν from a separate data source resolves this without imposing a single life-table assumption across the full sample.

21.2 Construction recipe: life-table on the LF

Data source. CPS Basic Monthly, same extract as for the transition moments (Section 2). Cross-section: no panel matching is required.

Sample.

1. Civilian, age 16–64.
2. In the labour force at date t (employed or unemployed, `LABFORCE` = 2).
3. Restricted to the window w being estimated ($w \in \{\text{base_fc}, \text{base_covid}\}$).

Remaining working life. For each LF observation (i, t) , the implied months remaining until age 65 are

$$\text{rem}_{i,t} = \max(65 - \text{AGE}_{i,t}, 0) \times 12.$$

The weighted mean over window w is

$$\overline{\text{rem}}_w = \frac{\sum_{(i,t) \in w} w_{i,t} \text{rem}_{i,t}}{\sum_{(i,t) \in w} w_{i,t}},$$

and the life-table turnover rate is the reciprocal:

$$\hat{\nu}_w = \frac{1}{\overline{\text{rem}}_w}.$$

Two values, one per crisis pair. We compute $\hat{\nu}_w$ for $w \in \{\text{base_fc}, \text{base_covid}\}$, and use the FC value $\hat{\nu}_{\text{base_fc}}$ for the (`base_fc`, `crisis_fc`) pair and the COVID value $\hat{\nu}_{\text{base_covid}}$ for the (`base_covid`, `crisis_covid`) pair. Allowing the two pairs to have distinct $\hat{\nu}$ values accommodates secular drift in the working-age composition of the LF between the early 2000s and the late 2010s (the LF has aged, lengthening $\overline{\text{rem}}_w$ slightly between the two baselines).

Why baseline windows. The pre-FC and pre-COVID baseline expansions are the “normal” regimes within each crisis pair. Computing the life-table within the baseline windows avoids contamination from crisis-period sample-composition shifts (e.g. disproportionate exit of older workers during downturns) and keeps the macro ν aligned with the data the SMM treats as the long-run regime for each pair.

Relation to the model. In the model, workers exit at Poisson rate ν regardless of labour-force status, and each exit is replaced by a newborn entrant drawn from $\ell(x)$. The life-table value $1/\overline{\text{rem}}_w$ is the right level for this: $\overline{\text{rem}}_w$ is the average months until forced retirement at age 65, so $\hat{\nu}_w$ is the average monthly hazard needed to cycle every worker out by then.